Striped bass (*Morone saxatilis*)

Summary Scores

- **Sensitivity:** High
- **Exposure:** Moderate
- **Vulnerability:** Moderate

Sensitivity

Sensitivity Attributes	Score	LH DQ	Tally
Adult Mobility	1.48	2.78	
Complexity in Reproductive Strategy	3.36	2.98	
Dispersal of Early Life Stages	3.08	2.78	
Egg and Larval Survival and Settlement Requirements	3.68	2.94	
Historical Variability in Temperature	1.44	2.88	
Juvenile Habitat Requirements	2.96	2.7	
Other Stressors	3.36	2.98	
Population Growth Rate	3.2	2.84	
Prey Specificity	1.36	2.78	
Relationship to Calcified Organisms (due to impacts from Ocean Acidification)	1.4	2.5	
Spawning Cycle	3.36	2.98	
Stock Size/Status	3.12	2.88	
Total Sensitivity | Certainty	**High (3) | 0.85**		

Exposure

Exposure Factors	Mean	SD	Tally
Bottom Temperature	3.09	0.82	
Bottom Oxygen	2.82	0.91	
Bottom Salinity	2.86	0.63	
Bottom Aragonite Solubility	1.03	0.19	
Small zooplankton integrated biomass	1.57	0.58	
Medium zooplankton integrated biomass	1.32	0.43	
Large zooplankton integrated biomass	1.32	0.48	
Total Exposure	**Moderate (2.27)**	**0.87**	

Vulnerability

Vulnerability	Mean	SD	Tally
Total Vulnerability	**Moderate (6.81)**	**2.6**	

Low Moderate High Very High

Figure 1: Summary Sensitivity, Exposure, and Vulnerability Scores and Score Distributions for Striped bass. Exposure and Vulnerability scores are averaged spatially across the species distribution and within stocks if applicable (see Figure 5 and Figure 4). Life History Data Quality (LH DQ) and Standard Deviations (SD) are reported for senstivity and exposure/vulnerability scores, respectively.

Changes from Previous Assessment(s)

Table 1: Changes to Vulnerability, Exposure, and Sensitivity scores relative to Previous Assessments. If a citation for the previous assessment is not provided, it comes from Hare et al. (2016).

Metric	Previous Assessment(s)	Current Assessment	Change
Sensitivity	High	High	None
Exposure	Very High	Moderate	Decrease
Vulnerability	Very High	Moderate	Decrease
Directional Effect	Neutral	Negative	Decrease

Stocks

Striped bass is managed as a single stock or stock information was not considered for this iteration of the CVA. ## Score Descriptions:

Sensitivity:

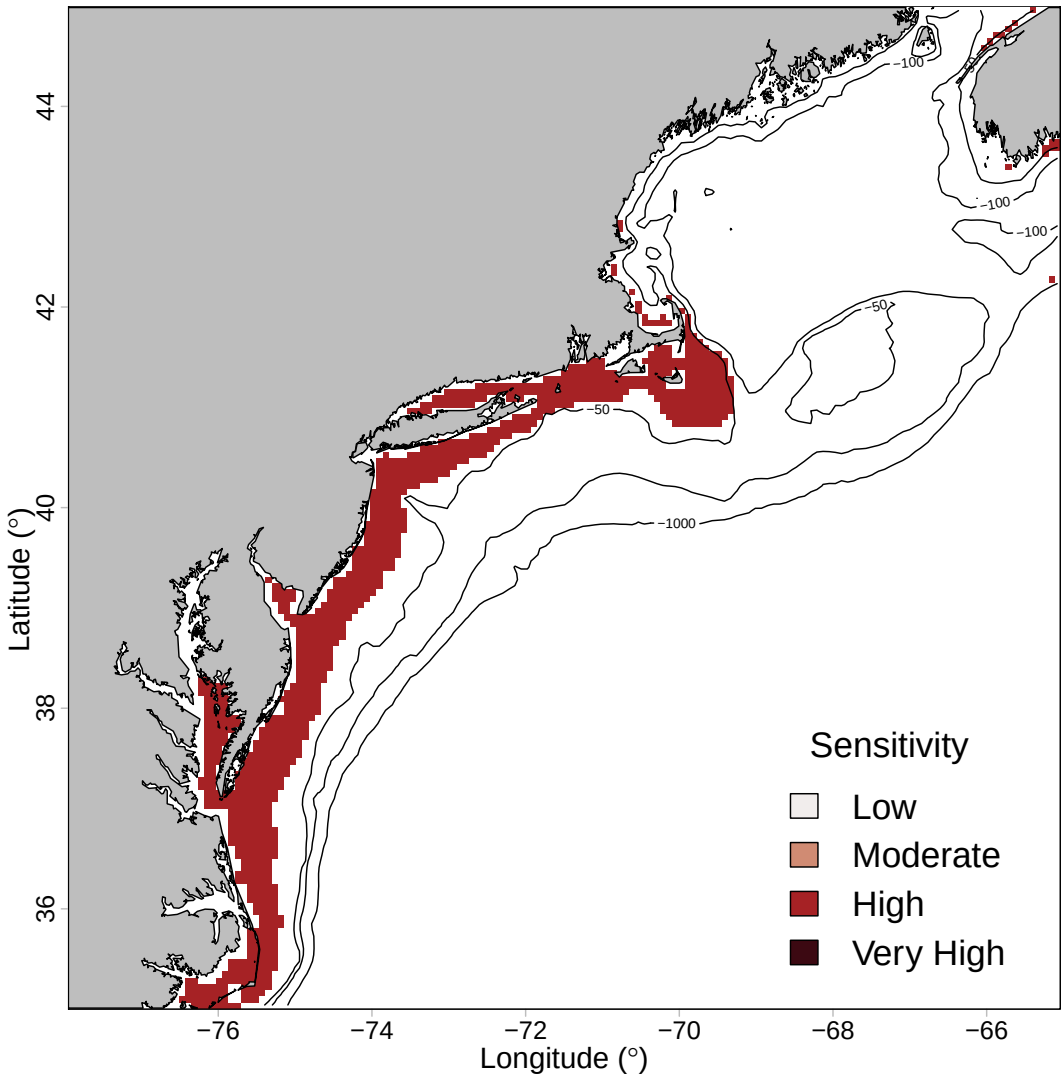


Figure 2: Striped bass Sensitivity on the Northeast Shelf

Striped Bass received a **High** sensitivity score. Certainty was 85%. A total of seven attributes had scores greater than 3: Egg and Larval Survival and Settlement Requirements (3.7), Complexity in Reproductive Strategy (3.4), Other Stressors (3.4), Spawning Cycle (3.4), Population Growth Rate (3.2), Stock Size (3.1), and Dispersal of Early Life Stages (3.1) (Figure 1).

Striped Bass are anadromous, spawning in fresh water once a year (Bigelow and Schroeder 1953). Estuarine and river water column habitats used by spawning adults are considered highly vulnerable (Guyant et al. 2022). Eggs and larvae are semi-buoyant drift downstream away from spawning sites and require relatively fast current speeds (0.3 m/s) to avoid sinking and being smothered (Fay et al. 1983). Predation and temperature play a large role in larval survival (Northeast Fisheries Science Center 2019) (Fay et al. 1983) (Smith and Kernehan 1981).

Striped bass are long lived, reaching maturity between 2-8 years of age and living up to 30 years (NEFSC 2019). The stock is currently overfished, but overfishing is not occurring (Atlantic States Marine Fisheries

Commission 2024).

In the previous Climate Vulnerability Assessment, striped bass received a High sensitivity score. Similar to our results, Spawning Cycle and Early Life History Requirements drove this score (Hare et al. 2016).

Exposure:

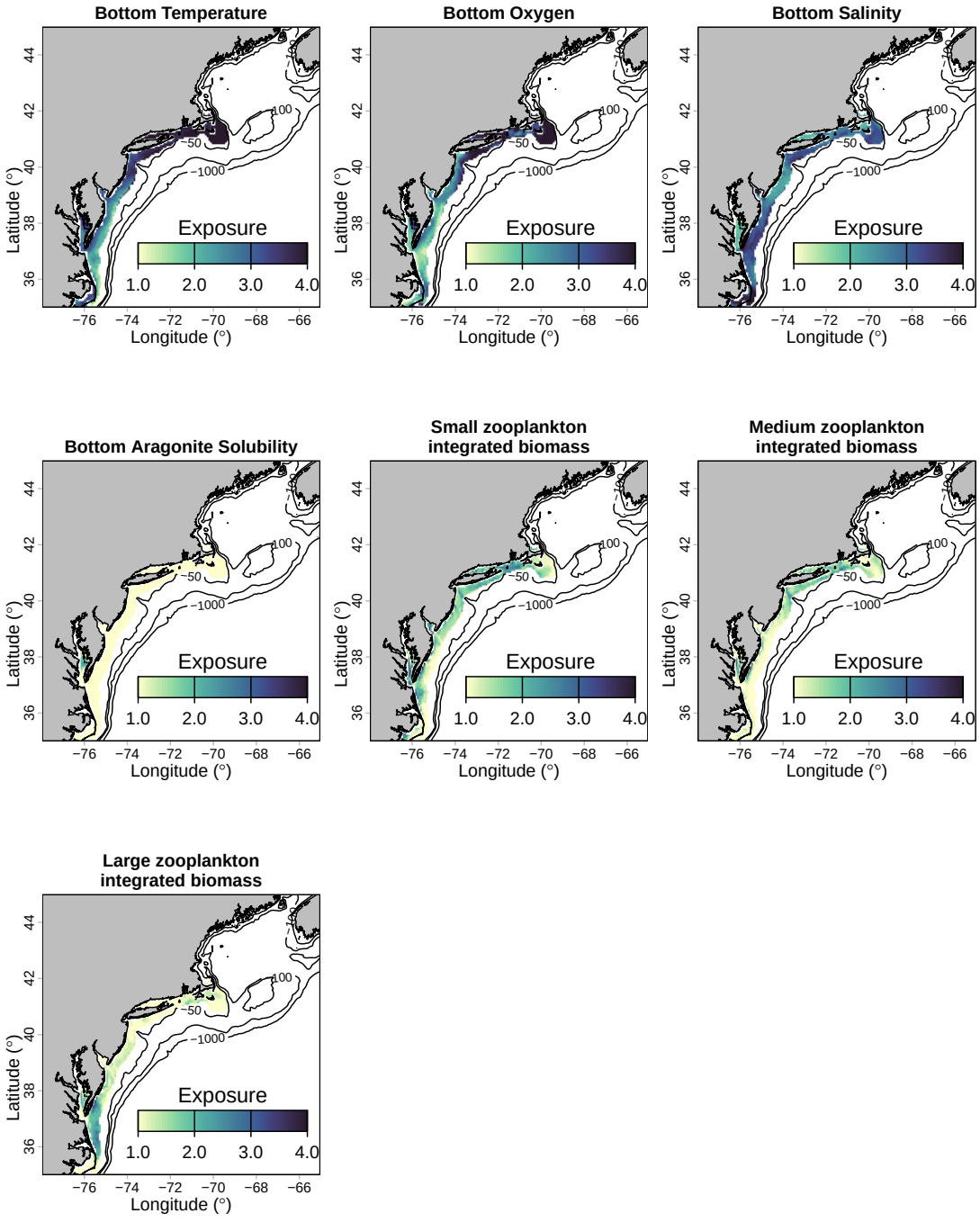


Figure 3: Maps of each Exposure Factor for Striped bass. Exposure was calculated between 2014-2023 and 2025-2035

Mean exposure for Striped Bass was **Moderate** (2.27 +/- 0.87) (Figure 1). Exposure was High in Southern New England, Long Island Sound, and Delaware and Chesapeake Bays. Exposure was lower along the

coasts from New Jersey to North Carolina (Figure 4). This was driven by high exposure scores for bottom temperature and bottom oxygen (Figure 3).

In the previous Assessment, Striped Bass received a Very High Exposure score, driven by very high sea surface temperature, air temperature, and ocean acidification (Hare et al. 2016).

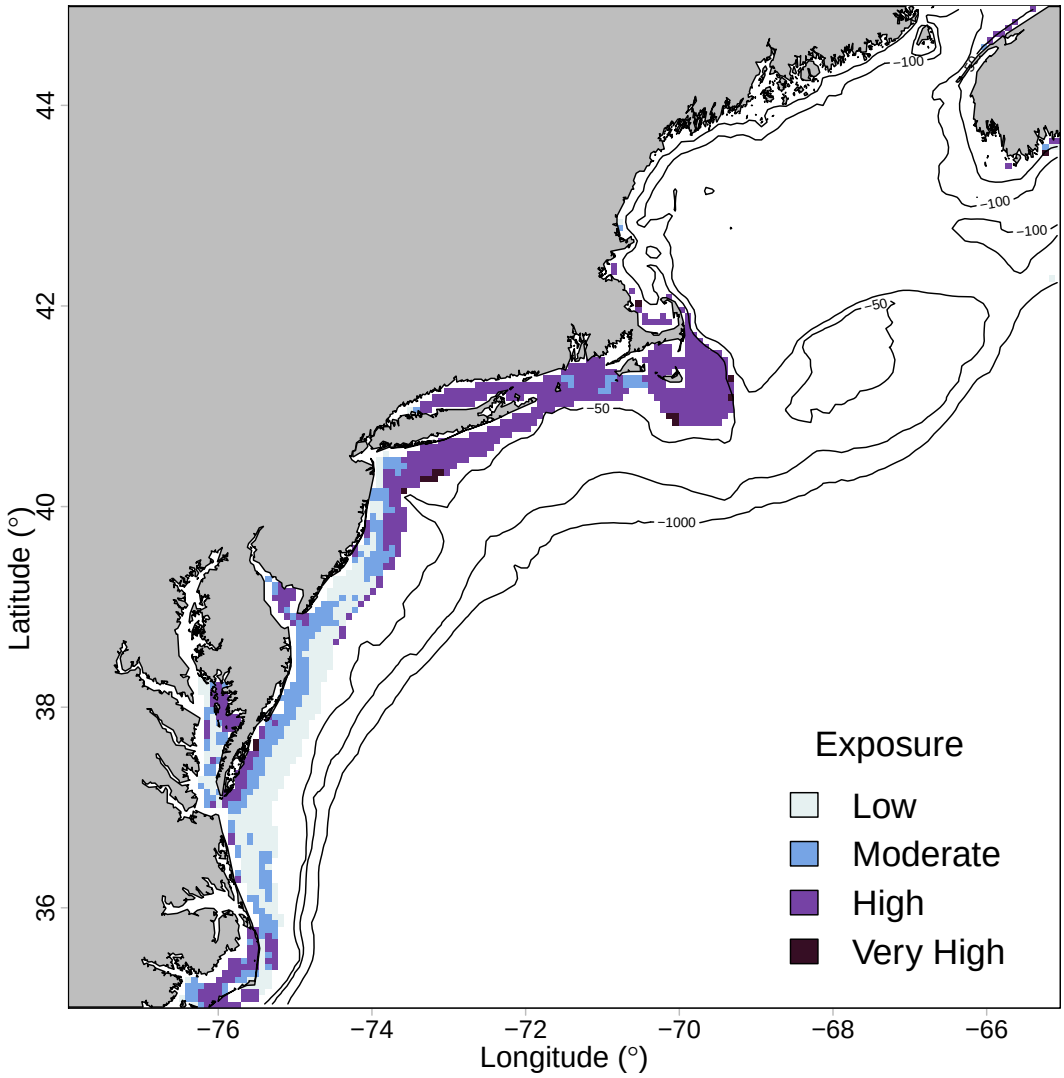


Figure 4: Total Exposure on the Northeast Shelf for Striped bass. Total Exposure is calculated by applying the logic rule from Hare et al. (2016) within grid cells across exposure factors (see Figure 3)

Vulnerability:

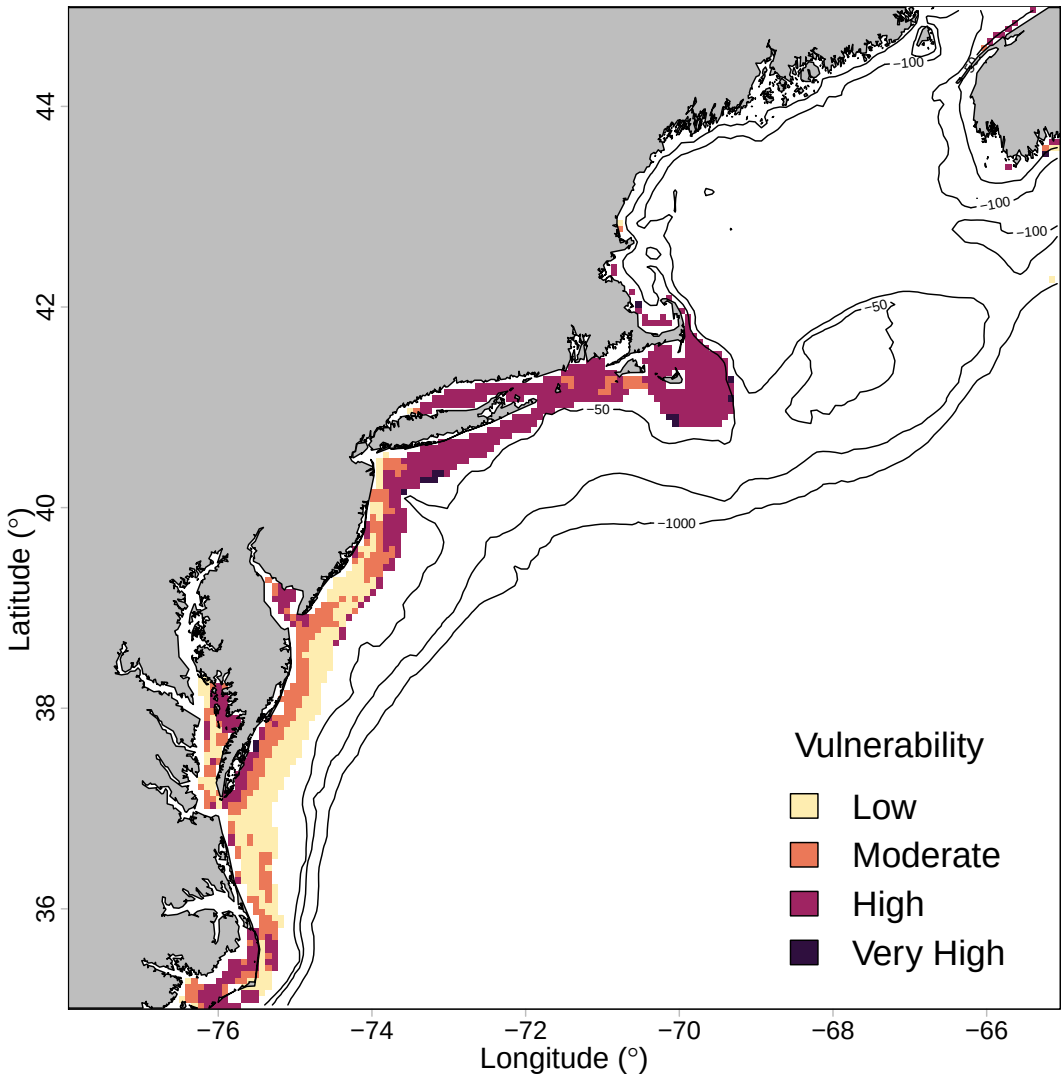


Figure 5: Vulnerability on the Northeast Shelf for Striped bass. Vulnerability is the product of sensitivity and exposure (see Figure 4 and Figure 2)

Overall mean Vulnerability for Striped Bass was **Moderate** (6.81 +/- 2.60) (Figure 1). Vulnerability was highest in Southern New England and in the major estuaries along the coast (Chesapeake Bay, Delaware Bay, Long Island Sound) (Figure 5).

In the previous Assessment, striped bass received a Very High Vulnerability score (Hare et al. 2016).

Data Quality Metrics

Life History Data Quality:

98% of data quality scores were greater than or equal to 2, indicating excellent data quality. No data gaps (mean attribute data quality score < 2) were identified (Figure 1).

Model Confidence Score:

The ensemble striped bass distribution model received a model confidence score of 1.7 (+/- 0.9), indicating low to moderate confidence in the model. The model predictions generally matched experts' expectations, however, they missed known migration patterns. This was attributed to catchability of striped bass in the survey datasets used. The ensemble model AUC score was 0.92, indicating excellent model predictive performance.

Additional Metrics

Distribution Shifts:

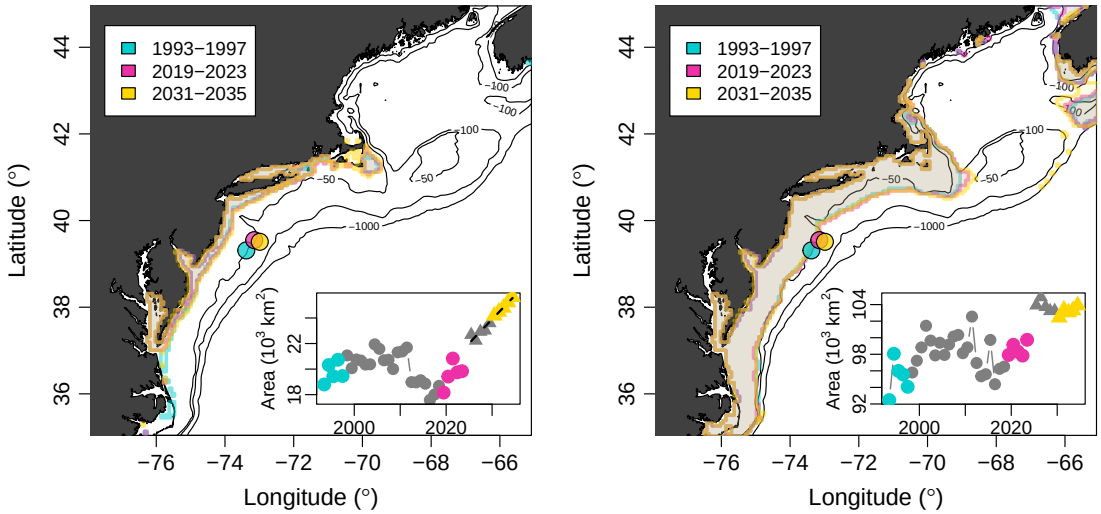


Figure 6: Distribution shifts on the Northeast Shelf for Striped bass. Polygons illustrate core habitat (left) and species range (right) for five-year average periods. Points illustrate weighted centroid, calculated as the average position with species distribution models as weights. Inset timeseries illustrate the shift in core and habitat range area over time. Circles represent points from the MOM6 hindcast; triangles represent points from the MOM6 forecast. Dashed lines represent significant linear regressions

Note: For an animated version of this plot, please see the online version of this document.

Striped bass core habitat and overall distributions did not shift significantly between 1993 and 2023 (Figure 6). The distribution centroid shifted 33 km northeast.

Under near future ocean conditions, striped sea bass habitat areas are predicted to expand. Striped bass distributions are expected to shift northward, including into Cape Cod Bay (Figure 6). As a result, the distribution centroid is projected to shift 15 km east.

In the previous assessment, striped bass received a High Distributional Vulnerability Rank due to their high mobility, large migrations, and variety of preferred habitats (Hare et al. 2016).

Table 2: Distribution shifts based on five-year average distributions. Changes are calculated relative to the previous time period.

Time Period	Weighted Centroid Shift (km & bearing)	Core Habitat Area (change) (km^2)	Habitat Total Area (change) (km^2)	Latitude Range (change) (deg)	Change in Longitude Range (deg)
1993-1997	NA	19,905	95,367	6.80	8.26
2019-2023	33.0 km NE	19,680 (-225)	98,747 (+3,380)	6.55 (-0.25)	8.34 (+0.08)
2031-2035	14.6 km E	24,881 (+5,201)	103,301 (+4,555)	6.85 (0.31)	10.18 (1.84)

Directional Effect:

Experts predicted that striped bass would be **negatively** impacted by climate change. Certainty amongst expert scores was 100%. Spawning and recruitment success is driven by freshwater outputs and subnormal winter temperatures (O'Connor et al. 2012) (Dixon et al. 2026) (Heinle et al. 1977). Estuarine and river spawning habitats are also highly vulnerable (Guyant et al. 2022). The previous Climate Vulnerability Assessment predicted that climate change would have a neutral effect on striped bass (Hare et al. 2016).

Climate Effects on Abundance and Distribution

Coutant (1990) analyzed habitat suitability for striped bass on the U.S. east coast with predicted temperature increases and declines in precipitation. In freshwater areas, high temperatures, altered streamflow, and low oxygen may constrain suitable habitat in the summer. Habitat loss in the Chesapeake Bay is predicted, with a possible range expansion into Nova Scotia and further into the Gulf of St. Lawrence (Coutant 1990). Onset and cessation of the spawning season in two spawning grounds of the Hudson River Estuary have occurred earlier in recent years. Spawning sites in the upper estuary are more vulnerable to climate change compared to sites in the lower estuary (Pan et al. 2023). Larval striped bass abundance in the Hudson River Estuary is positively correlated with high freshwater flows, which may decrease under predicted climate scenarios (O'Connor et al. 2012). Striped bass recruitment has been linked to winter North Atlantic Oscillation (NAO) and marine heatwaves (Dixon et al. 2026). Several studies support a general conclusion that dominant year classes through recent history have always been preceded by a colder than normal winter (i.e. subnormal winter temperatures). One proposed mechanism of the severe winter-strong year class association is through the estuarine food chain (Heinle and Flemer 1975) (Heinle et al. 1977).

Life History Synopsis

Atlantic striped bass inhabit freshwater coastal and estuarine waters from Canada to Florida, including the Gulf of America from Florida to Louisiana (Northeast Fisheries Science Center 2019). Striped bass are anadromous, spawning once a year in fresh or nearly fresh water. Along the Atlantic coast, the spawning period ranges from mid-February in Florida to June and July in the Gulf of St. Lawrence. In the mid-Atlantic, spawning is concentrated during the months of April - June (Bigelow and Schroeder 1953). Spawning peaks are apparently triggered by a noticeable increase in water temperature, generally beginning at temperatures of at least 14°C (Johnson and Koo 1975). Eggs hatch within 30 - 74 hrs, occurring faster at higher temperatures. Larval stages last up until age 1 when juveniles migrate out of

rivers and into estuaries (Northeast Fisheries Science Center 2019). Larvae and eggs may drift down stream in rivers, away from spawning sites. Eggs are semi-buoyant, requiring a current of 30 cm/sec in order to keep the eggs afloat so they don't settle to the bottom and die (Fay et al. 1983). Egg and larval mortality, as well as larval growth rates and yolk utilization are all linked to water temperatures greater than 18°C (Northeast Fisheries Science Center 2019). Predation and competition by American shad are important factors in spawning and larval locations (Fay et al. 1983). In addition, adult and juvenile white perch and copepod *Cyclops bicuspidatus* probably consume large numbers of striped bass larvae (Smith and Kernehan 1981). Larval striped bass feed on zooplankton; juveniles eat small shrimps and other crustaceans, annelid worms, bay anchovies, gobies, and insects (Bason 1971). Most juveniles remain in the river and estuarine areas where they were spawned. During the first summer, young-of-the-year striped bass generally move downstream and shoreward (Fay et al. 1983). A variety of environmental factors influence the timing of this migration, including temperature, pH, and salinity (Northeast Fisheries Science Center 2019). Males reach maturity at age 2-4 and females from age 4-8. Striped bass have a fairly long lifespan, up to 30 years (Northeast Fisheries Science Center 2019). Adults are generalists, consuming a variety of fish and invertebrates, depending on fish size, time of year, and habitat (Northeast Fisheries Science Center 2019). Adults have few predators, with the exception of some seals and sharks (Northeast Fisheries Science Center 2019). The Atlantic States Marine Fisheries Commission manages striped bass in coastal and estuarine waters from Maine to Virginia, and coastal areas of North Carolina under Amendment 7 of the Interstate Fishery Management Plan (Atlantic States Marine Fisheries Commission 2022). Estuarine waters of North Carolina are managed by the state. The stock is currently overfished, with management measures and a rebuilding plan in place, but not subject to overfishing (Atlantic States Marine Fisheries Commission 2024).

Supplemental Scores

Exposure Timeseries

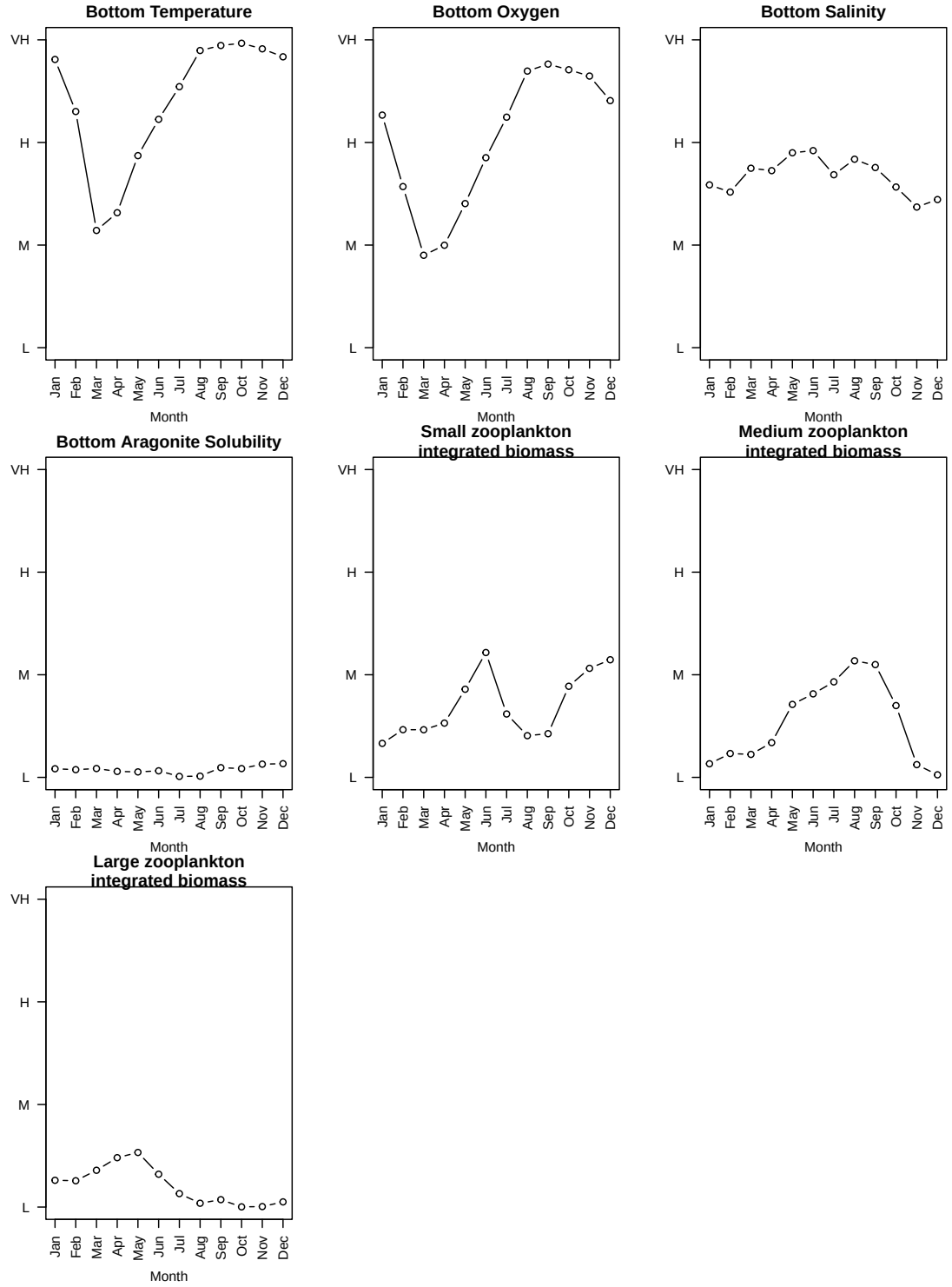


Figure 7: Monthly Timeseries of each Exposure Factor for Striped bass. Exposure was calculated between 2014-2023 and 2025-2035

Across the year, Exposure ranged from Low to High for Striped Bass. Scores were lowest in March and April and Moderate to High the rest of the year (Figure 8). This was driven by declines in bottom temperature and oxygen exposure scores (Figure 7).

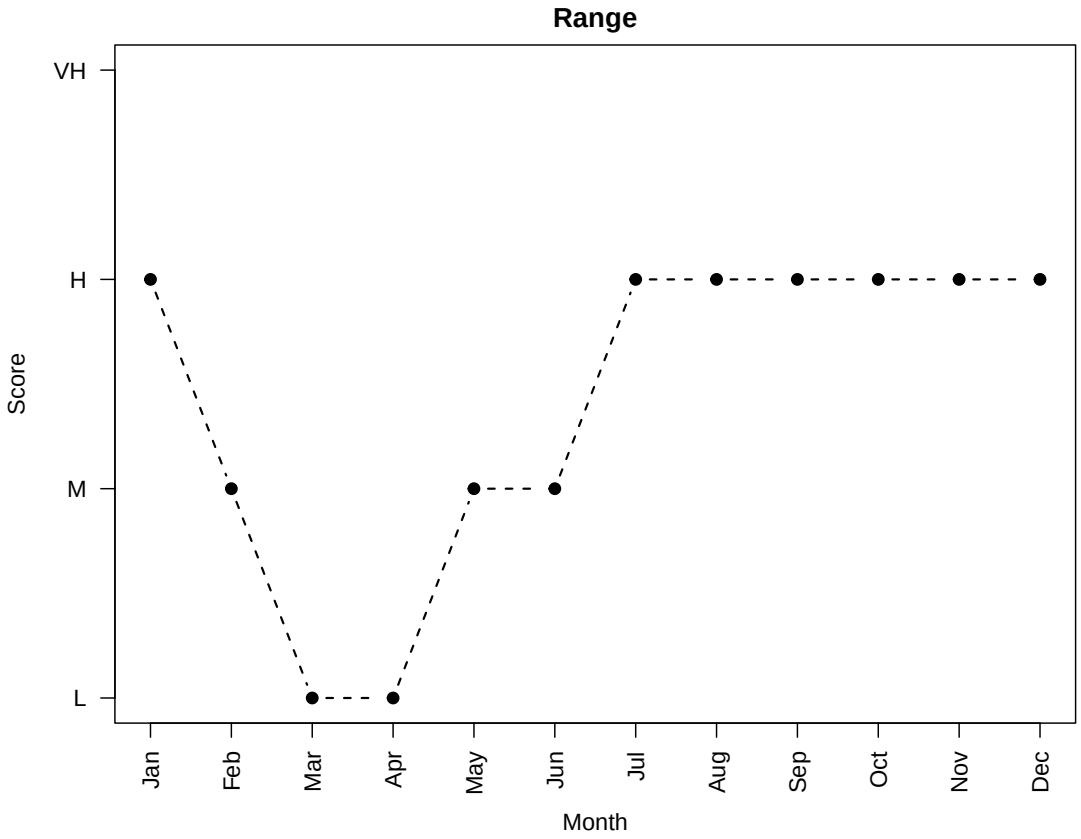


Figure 8: Timeseries of Total Exposure for Striped bass

Exposure due to Significant Factors

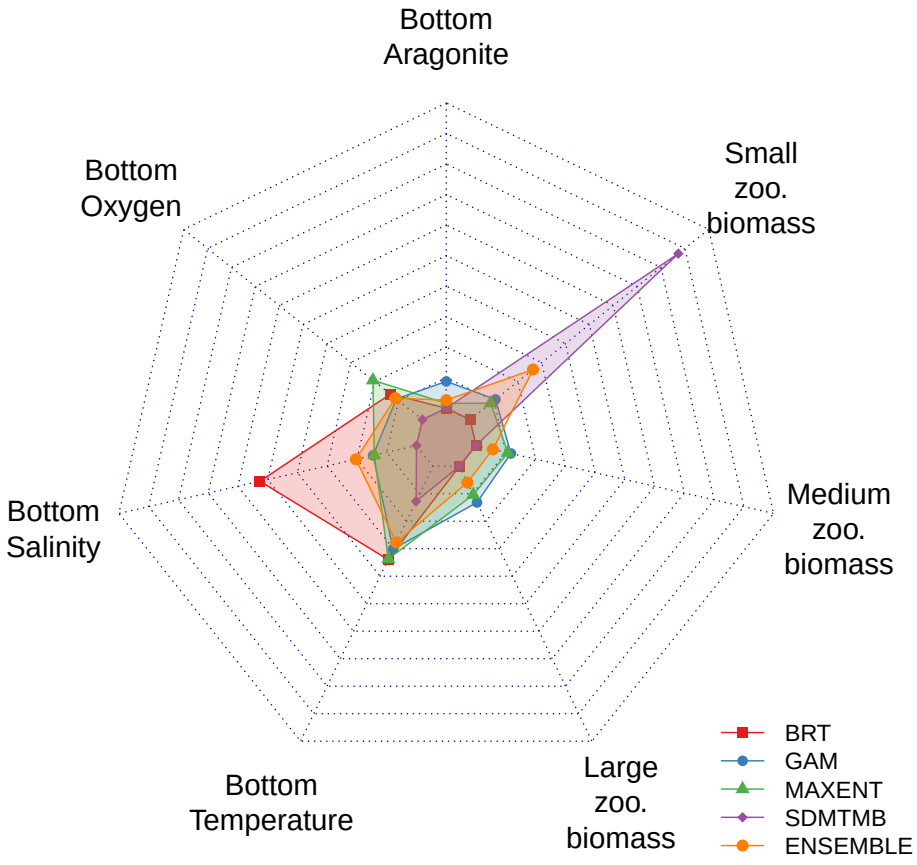


Figure 9: Dynamic Factors driving Striped bass distributions. The closer the polygon is to the edge of the figure, the more important the variable is. Each ring represents 10% variance explained. Factors must explain at least 10% of variance in the ensemble model to be used to calculate total exposure.

Bottom temperature, oxygen, and salinity, in addition to small zooplankton biomass were important dynamic drivers of striped bass distributions (Figure 9). Striped bass inhabit estuarine and freshwater ecosystems and are generalist predators of smaller fish and invertebrates (Northeast Fisheries Science Center 2019). When only these factors were used to calculate exposure, scores remained Moderate on average (2.25 +/- 0.88) (Figure 11; Figure 10).

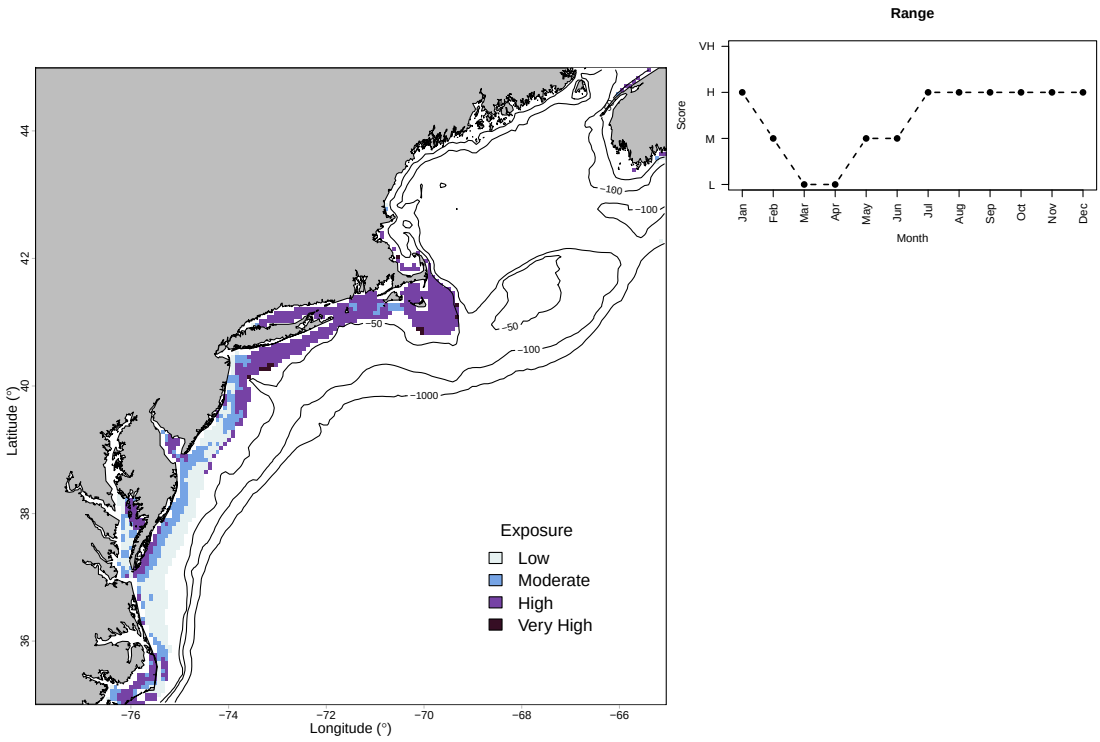
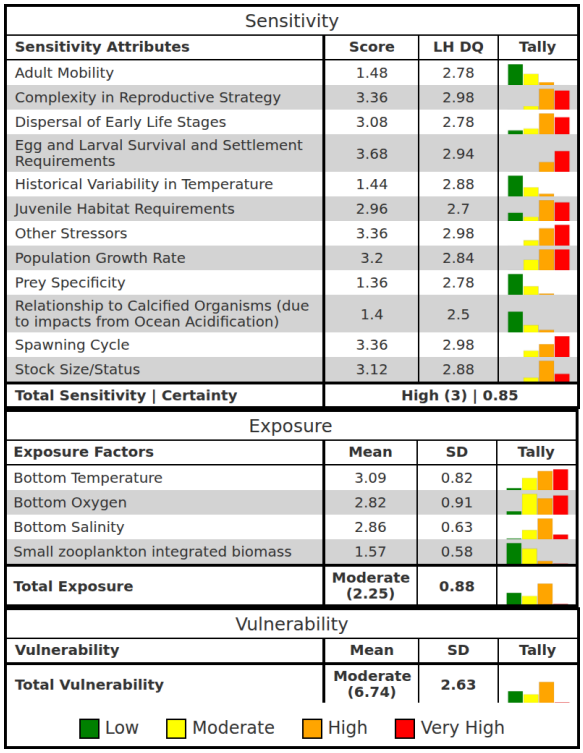


Figure 10: Total Exposure Map and Timeseries when Exposure was calculated with only Exposure Factors critical to Striped bass distributions



Sensitivity

Sensitivity Attributes	Score	LH DQ	Tally
Adult Mobility	1.48	2.78	
Complexity in Reproductive Strategy	3.36	2.98	
Dispersal of Early Life Stages	3.08	2.78	
Egg and Larval Survival and Settlement Requirements	3.68	2.94	
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Spawning Cycle	3.36	2.98	
Stock Size/Status	3.12	2.88	
Total Sensitivity | Certainty	**High (3) | 0.85**		

Exposure

Exposure Factors	Mean	SD	Tally
Bottom Temperature	3.09	0.82	
Bottom Oxygen	2.82	0.91	
Bottom Salinity	2.86	0.63	
Small zooplankton integrated biomass	1.57	0.58	
Total Exposure	**Moderate (2.25)**	**0.88**	

Vulnerability

Vulnerability	Mean	SD	Tally
Total Vulnerability	**Moderate (6.74)**	**2.63**	

Low Moderate High Very High

Figure 11: As in Figure 1 for Striped bass, but with only significant Exposure factors and the resulting Vulnerability Scores.

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